VoiceOps 2026: Capstone Project Report

LLM & Generative AI Course · Consulting & Analytics Club, IIT Guwahati × Gnani.ai
Project: End-to-End Multimodal AI Customer Support Agent for ZentroMart
Reference Date: 20 October 2026

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1. Executive Summary

This report documents the architectural design, implementation, and empirical evaluation of **VoiceO
ps 2026**, an end-to-end multimodal customer support agent for **ZentroMart**, a fictional Indian on

The agent guides customers across text and spoken channels, operating in four sequential stages:
1. **Step 1 (Prompting LLM):** System prompting establishing AI assistant identity and behavioral bo
undaries.
2. **Step 2 (RAG Grounding):** Document chunking, precedence weighting, Section ID tracking, and str
ict hallucination deterrence.
3. **Step 3 (Conversational Intelligence & Verification):** Multi-turn context windows, strict two-s
tep identity verification gating, mock tool integration (`lookup_order`, `create_ticket`), and escal
4. **Step 4 (VoiceOps Integration):** Speech-to-Text (STT) and Text-to-Speech (TTS) integration usin
g **Gnani.ai** models (`timbre-v2.5`), text normalization for speech, and batch execution capabiliti

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2. Architecture & Step-by-Step Implementation

Step 1: Prompting Baseline & Assistant Persona

In Step 1, the agent was configured with a comprehensive system prompt establishing its role as Zent
roMart's AI support assistant. As mandated by `agent_conversation_style.md` [STY-1], the agent's ver

In baseline mode (prompting only), the LLM lacks access to internal policy documents and catalog rec
ords, relying purely on parametric memory. This established a quantitative baseline for measuring ha

Step 2: Knowledge Base Indexing & Hybrid RAG Engine

To eliminate hallucinations and ground answers in company truth, we constructed a high-precision RAG
 engine indexing all 21 company documents in `kb/` and 70 products in `data/products.csv`:

- **Granular Chunking:** Markdown documents were segmented by section headers (`## [SECTION-ID]
  Section Title`). Each chunk retained its metadata (`doc_id`, `section_id`, `doc_type
- **Product Catalog Indexing:** Every catalog item was indexed under the standardized key `prod
  ucts.csv:{sku}` with brand, category, price, and ZentroMart Assured status.
- **Precedence Hierarchy Enforcement (`KB-00.1`):** A strict multiplier was implemented priorit
  izing:

$$\text{Current Policy (1.4)} > \text{Agent Guides (1.25)} > \text{Help (1.15)} > \text{FAQ (1.05)}
> \text{Marketing (0.85)}$$

Superseded policies (such as `pol_returns_v3_superseded.md`) receive a heavy penalty (0.35) unless t
he query explicitly concerns orders delivered before 1 August 2026.

- **Strict Out-of-Scope Handling:** When retrieved confidence falls below threshold or queries
  target out-of-scope policies (e.g., midnight delivery, GST invoices for bulk orders,

Step 3: Multi-Turn Conversation, Identity Verification & Tool Integration

The conversational engine implements a state machine maintaining dialogue state across multiple cust
omer turns:

- **Identity Verification Gating (`agent_identity_verification.md`):**
- General policy queries require no caller verification.
- For order-specific lookups, the agent invokes `lookup_order(order_id)`.
- Details remain strictly withheld until the caller supplies ONE valid proof: the **last 4 digi
  ts of the registered phone** OR the **delivery pincode**.
- A maximum of **one retry** is permitted upon mismatch. If verification fails twice, order dis
  closure is refused.
- **Date & Calendar Rules:**
- Business days run **Monday to Saturday**, skipping the 4 warehouse holidays in `data/holidays
  .csv` (15 Aug, 02 Oct, 19 Oct, 09 Nov 2026).
- Return windows use calendar days from delivery date (e.g., 7 days for electronics delivered a
  fter 1 Aug 2026; 30 days for Festive Sale fashion orders placed between 1·15 Oct 2026
- **Human Escalation Protocol (`agent_escalation_handoff.md`):**
- Triggers: Claims for damaged items $> ·10,000$ (Priority Resolution), suspected fraud, repeat
  contacts (≥ 2 prior calls), safety hazards, and overdue refunds past max busines
- Automatically invokes `create_ticket(summary=...)` with structured summaries ≥ 20 charact
  ers before confirming handover.

Step 4: Voice Integration via Gnani.ai Models

The voice layer bridges speech and conversational intelligence:

- **Gnani STT:** Transcribes customer audio (supports `en-IN`, `hi-IN`, `hi-en`, and regional l
  anguages) with recovery from silence/unclear audio.
- **Ear Text Normalization:** Formats text for natural speech: spaces out 8-digit order numbers
  (`8 9 7 8, 2 8 3 7`), spells out rupee currency terms (`49 rupees`), and expresses d
- **Gnani TTS:** Synthesizes voice audio using model `timbre-v2.5` and voice `Kaveri`.
- **Batch Evaluation Runner (`run.py`):** Accepts an input directory of customer audio recordin
  gs, generates transcripts, executes conversational logic, retrieves chunks, and saves

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3. Evaluation Results: Step 1 vs. Step 2 vs. Step 3

The system was evaluated against the 52 questions in `practice/questions.csv` and the 23 conversatio
ns (96 customer turns) in `practice/conversations.csv`:

Step 1 vs. Step 2 Performance Comparison

Evaluation Metric | Step 1 (Prompting Baseline) | Step 2 (RAG Grounded) | Delta / Impact
Turn Decision Accuracy | 88.46% | **88.46%** | Grounded consistency
Gold Evidence Section Recall | 0.00% | **97.87%** | $+97.87\%$ (Accurate retrieval)
Hallucination on Unanswerable Queries | 60.00% | **0.00%** | **-60.00% (Hallucina
Unanswerable Policy Detection | 0.00% | **100.00%** | $+100.00\%$ (Polite boundary)
Must-Not-Include Violations | 4 violations | **0 violations** | Zero policy leaks

Step 3: Multi-Turn Conversation & Tool Execution (96 Turns)

Multi-Turn Metric | Evaluated Score | Benchmark Requirement
Conversation Decision Accuracy | **100.00% (96/96)** | Exceeds expected benchmark
Tool Call Invocation Accuracy | **100.00% (96/96)** | Accurate `lookup_order` & `create_

Verification Gate Enforcement | **100.00%** | Zero private data leakage before proof
Must-Include Dialogue Score | **92.71% (89/96)** | High conversational adherence
Escalation Protocol Compliance | **100.00%** | Valid summaries generated

Key Learnings

1. **Parametric Hallucination vs Grounded Truth:** Without RAG, LLMs hallucinate non-existent studen
t discounts and store pickup options. Grounded retrieval with explicit unanswerability thresholds co
2. **Precedence Resolution:** Real customer policies conflict (e.g. general return FAQs stating 10 d
ays vs current v4 policy establishing 7 days for electronics). Applying precedence weights ensured c
3. **Identity-Checked Safety:** Stateless tools require stateful conversation managers. The session
state machine successfully tracked verification attempts, preventing brute-force disclosures while p

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4. Challenges & Engineering Solutions

1. **ASR Dialect & Multilingual Variances:**
   · *Challenge:* Practice questions contained phonetic ASR variations (e.g., "ele-zero zero one t
   o", "aakhri char ank", "kitne din me wapas").
   · *Solution:* Implemented dual lexical-semantic matching with regex token normalization mapping
   Hindi/Hinglish colloquialisms to canonical catalog and policy concepts.
2. **Business Day & Holiday Arithmetic:**
   · *Challenge:* Calculating whether a refund is overdue required navigating Monday·Saturday busi
   ness days while skipping specific warehouse closure holidays (e.g., Diwali / festival
   · *Solution:* Built a calendar utility integrating `data/holidays.csv` that precisely computes
   dates (e.g., mapping an initiation on Saturday 17 October to 27 October 2026).
3. **Voice Ear Normalization:**
   · *Challenge:* Raw LLM markdown output (asterisks, bullet points, raw 8-digit numbers) sounds r
   obotic and confusing over TTS.
   · *Solution:* Developed a speech normalizer converting `·` to `rupees`, expanding dates into sp
   oken prose, and inserting space pauses into order IDs.

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5. Bonus Deliverables & Enhancements

· **Interactive Streamlit Web Interface (`app.py`):** Live voice recording, audio playback, ord
er lookup simulator, escalation ticket log, and real-time telemetry inspector.
· **Dual Cloud & Offline Resilience:** Fully integrated with Google Gemini and Groq API keys, w
ith automatic deterministic fallback ensuring reproducible execution in offline evalu
· **Standardized Submission Structure:** Complete with `run.py`, `run.sh`, `requirements.txt`,
`.env.example`, and comprehensive documentation.